27. The minimum intensity of light that the human eye can detect is 2.0×10^{-11} W m^{-2}.

Given that the pupil of the eye has a diameter of 4.0 mm, how many photons per second of wavelength 550 nm must enter the eye for a distant star to be visible?

A 700 **B** 2800 **C** 1.3×10^9 **D** 5.1×10^9

Ans: A

$$\text{Minimum intensity } I = \frac{\textit{Minimum power received by eye}}{\textit{Area of pupil}}$$

$$= \frac{(\textit{Rate of photon arrival})(\textit{Energy of each photon})}{\pi r^2}$$

$$= \frac{\frac{N}{t} h f}{\pi r^2} = \frac{\frac{N}{t} h \frac{c}{\lambda}}{\pi r^2}$$

$$2.0 \times 10^{-11} = \frac{\frac{N}{t}\left(6.63 \times 10^{-34}\right)\frac{3.00 \times 10^{8}}{550 \times 10^{-9}}}{\pi (0.002)^2}$$

$$\frac{N}{t} = 700 \text{ (2sf)}$$

28. In the figure below, graph P shows an X-ray spectrum from an X-ray tube